

MAHARAJA SURAJMAL INSTITUTE OF TECHNOLOGY

Department of Information Technology

6th semester

Subject: Advanced Java Programming Code: CIE306T

MINI PROJECT

Distributed E-Commerce System

Project Statement

This project aims to develop a multi-tier distributed e-commerce application using JavaBeans, JSP/Servlets, Hibernate, and RMI. The system enables users to browse products, place orders, and perform secure transactions, while administrators manage inventory and analytics.

System Architecture

- Presentation Layer: JSP + Servlets
- Business Layer: JavaBeans
- Persistence Layer: Hibernate
- Distributed Layer: RMI

Modules

- User Module: Registration, Login, Browse Products, Add to Cart, Place Order
- Product Module: Listing, Search, Category Management
- Payment Module (RMI): Remote payment processing and validation
- Order Management: History, Tracking, Invoice
- Admin Module: Manage Users, Products, Reports

Technology Usage

- JSP/Servlets for MVC-based handling
- JavaBeans for business logic
- Hibernate for ORM and database interaction
- RMI for distributed services

Database Design

- Users (UserID, Name, Email, Password)
- Products (ProductID, Name, Price, Stock)
- Orders (OrderID, UserID, Date, Status)
- Cart (CartID, UserID, ProductID, Quantity)

Advanced Features

- Session Management

- Form Validation
- Exception Handling
- Logging
- Secure Authentication
- Distributed Processing via RMI

Expected Outcome

A fully functional enterprise-level application demonstrating multi-tier and distributed architecture.